IS STRATEGY, MANAGEMENT AND ACQUISITION

CASE STUDY:

LOST AND FOUND MANAGEMENT SYSTEM

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INTRODUCTION

In many public schools in the Philippines, including **Pitos National High School**, the process of handling lost and found items remains largely manual. Students and teachers who misplace their belongings often rely on verbal reports, handwritten logs, or bulletin board announcements posted by the school administration. While this traditional approach has been used for years, it leads to several recurring problems such as unclaimed items, inaccurate records, and difficulty validating rightful ownership.

As the number of students continues to grow and school activities become more dynamic, misplaced items such as IDs, notebooks, wallets, uniforms, tumblers, and gadgets occur more frequently. Without a centralized system to track these items, many remain unclaimed and end up being stored indefinitely or discarded. This lack of documentation also makes it challenging for school administrators to verify claims, monitor patterns, or ensure transparency in item handling.

Recognizing the need for a more organized and efficient process, this case study proposes the development of a **Lost and Found Management System** specifically tailored for Pitos National High School. The system aims to digitize reporting, storage, and retrieval of lost or found items through a centralized platform accessible to both students and administrators. Through this initiative, the school can improve item recovery rates, reduce administrative workload, and establish a structured, transparent, and secure method for managing lost and found operations.

Statement of the Problem

The current lost and found process at Pitos National High School is handled manually through verbal reports, physical logs, and bulletin board postings. This outdated system leads to multiple operational issues that affect both students and school personnel. The absence of a centralized digital platform creates inefficiencies and increases the likelihood of miscommunication, unclaimed items, and inaccurate documentation.

This study aims to address the following problems:

1. Lack of a centralized system for recording and managing lost and found items, resulting in scattered and incomplete information.
2. Low item recovery rate because students and teachers have no easy way to check if their lost belongings have been found.
3. Difficulty verifying ownership due to missing documentation and lack of digital proof submission.
4. Inconsistent and inaccurate manual records, making it hard to track items, claims, and item history.
5. Delayed communication between students and administrators regarding found items or claim updates.
6. No automated alerts for items that remain unclaimed for long periods, leading to storage inefficiency or loss.
7. Lack of transparency, which may result in disputes or confusion over the rightful owner of an item.

Objectives of the Study

The primary purpose of this study is to design and propose a **Lost and Found Management System** for Pitos National High School that will streamline the reporting, verification, and retrieval of lost and found items. This system intends to improve efficiency, accuracy, and transparency in the school's item recovery process.

To develop a Cloud Based Lost and Found Management System that enhances item tracking, reporting, verification, and claim management within Pitos National High School.

1. To provide an online platform where students and teachers can easily report lost or found items.
2. To implement a digital verification and approval process to ensure proper identification of rightful owners.
3. To centralize all lost and found data into a single database for easy tracking and monitoring.
4. To enable search and filter functions for quick identification of items.
5. To automate notifications for claim updates, status tracking, and unclaimed item alerts.
6. To enhance the security and integrity of records through user authentication and access controls.

Benefits

The proposed Cloud-Based Lost and Found Management System for Pitos National High School provides several benefits to students, teachers, and administrators:

For Students & Teachers

- Faster Item Recovery – Users can quickly search if their lost item has been found.
- Convenience – Reports can be submitted anytime using a phone or computer.
- Improved Transparency – Users can track item status (Found → To Be Claimed → Successfully Claimed).
- Better Communication – Automatic notifications provide updates about claims or matches.

For School Administrators

- Organized Record Keeping – All lost and found items are stored in a centralized cloud database.
- Easy Verification – Allows quick checking of submitted photos or details to confirm ownership.
- Reduced Workload – Automated notifications and digital logs minimize manual tracking.
- Analytics for Decision-Making – Admin can generate reports on common items lost, peak periods, etc.

Overall School Benefits

- Enhanced Security & Accountability – Clear logs prevent disputes and ensure transparency.
- Efficient Management – Items are processed systematically, reducing clutter and misplacement.
- Modernized School Services – Supports the school's digital transformation initiatives.

System Functions

The system is designed to efficiently manage lost and found items by providing a comprehensive set of functions that streamline reporting, tracking, verification, and claim processes. Each function is tailored to enhance user experience, maintain data integrity, and ensure secure and organized management of items.

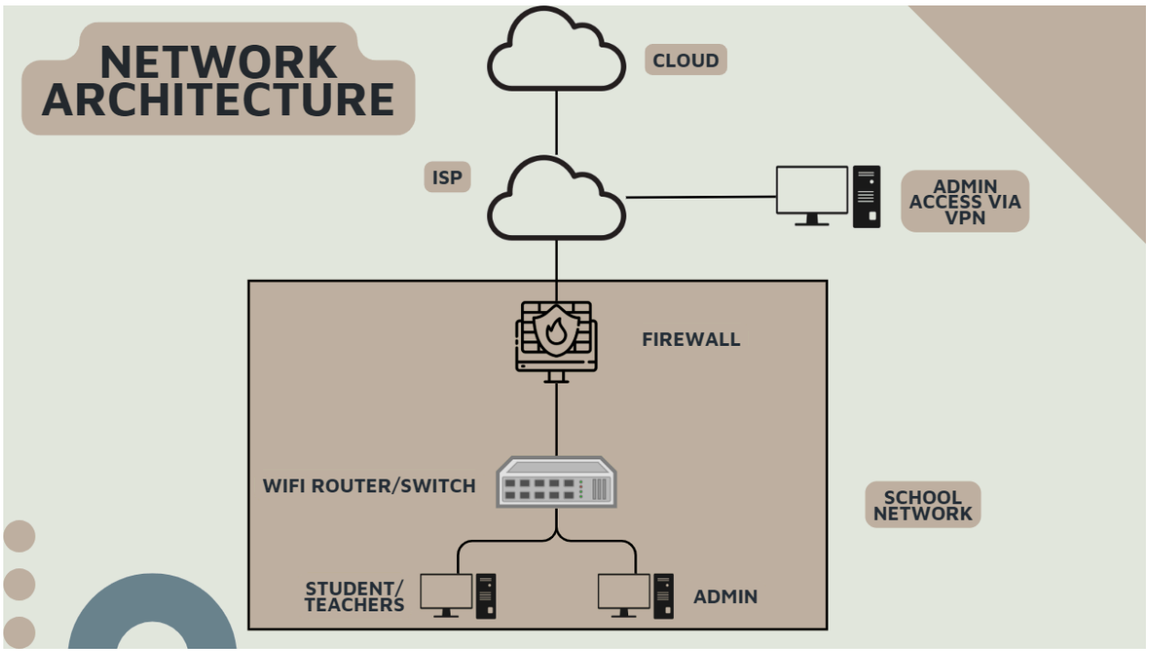
The key functions of the system include:

- **Lost Item Reporting** – Allows users to report lost items promptly, ensuring the system keeps a record for future reference.
- **Found Item Reporting** – Enables users to submit information about items they have found, contributing to a centralized database.
- **Item Verification & Approval** – Admins verify the authenticity of reported items and approve them for claim or storage.
- **Claim Management** – Facilitates the process of claiming items for users once verification is complete.
- **Notifications** – Sends timely alerts to users and admins about updates, approvals, or pending claims.
- **Centralized Item Database** – Maintains all records of lost and found items in a single, organized repository.
- **Search & Filter Options** – Provides users with tools to efficiently locate items using specific criteria such as name, date, or category..
- **System Security** – Ensures that all data is protected from unauthorized access, maintaining privacy and integrity.
- **Status Tracking** – Monitors the progress of lost and found items from reporting to claim completion.

Together, these functions create a robust and reliable system that simplifies the management of lost and found items, improves user convenience, and supports effective administrative oversight.

N**etwork Architecture**

Network architecture is the design of how devices and systems are connected to communicate and share data. For this system, it shows how computers, mobile devices, and cloud servers work together to manage lost and found items efficiently and securely. A clear network architecture ensures smooth operation, reliable data access, and proper communication between all components.



IT Equipment Materials

IT Equipment / Materials	
Wifi Router	Provides wireless internet connectivity for devices
UTP Cable	Connects devices to the network for stable wired communication
RJ45	Connector for UTP cables to link devices to network
Mobile	Used by staff to access system remotely
Laptop/ PC	Used for system management, reporting, and administration
VPN	Ensures secure remote access to the network
Wifi Dongle	Provides wireless connectivity to devices without built-in WiFi
Anti Virus Software	Protects devices from malware and cyber threats

Cost Benefit Analysis

A Cost-Benefit Analysis evaluates the total costs of implementing a system against the expected benefits, both tangible (quantifiable) and intangible (non-quantifiable). It helps determine whether the investment is justified, highlights potential savings, and shows how improvements in efficiency, productivity, and satisfaction outweigh the initial and recurring expenses.

ITEM	COST(PHP)
Wifi Router/Switch	15,000
Firewall	25,000
UTP Cable & RJ45	2,000
Wifi Dongle	3,000
ISP Subscription(monthly)	1,999

Cloud Hosting	1,999
SSL Certificate	3,400
VPN Service	1,100
Anti-Virus & Endpoint Security	5,000
PC, MOBILE, LAPTOP	40,000
Training & Documentation	5,000
Support & Maintenance	10,000
TOTAL	106,500 PHP

TANGIBLE BENEFITS		
Reduced admin workload	Saves 1–2 hours/day of staff time	₱3,000/month
Lower storage costs	Fewer unclaimed items reduce space requirements	₱1,000/month
Improved item recovery	Increase in items returned to owners (50–80%)	Reduces losses
Reduced paperwork	Digital logs replace paper registers	₱500/month saved
Faster claim process	Time saved per student in claiming items	₱500/month equivalent

INTANGIBLE BENEFITS	
Faster reporting & claiming	Improves satisfaction for students and staff
Higher transparency & accountability	Clear tracking of items prevents errors and disputes
Organized record-keeping	Easy retrieval
Support for digital transformation	Aligns with school modernization goals

The proposed system, with an initial cost of approx. ₱106,500, provides both tangible and intangible benefits. Tangible benefits like reduced admin workload, faster item recovery, and lower storage costs generate measurable savings of around ₱5,000 per month, allowing the system to pay for itself within 21–27 months. Intangible benefits, including improved transparency, student satisfaction, and staff morale, further enhance the value of the investment, making it highly advantageous for the school's operations.

SWOT Analysis

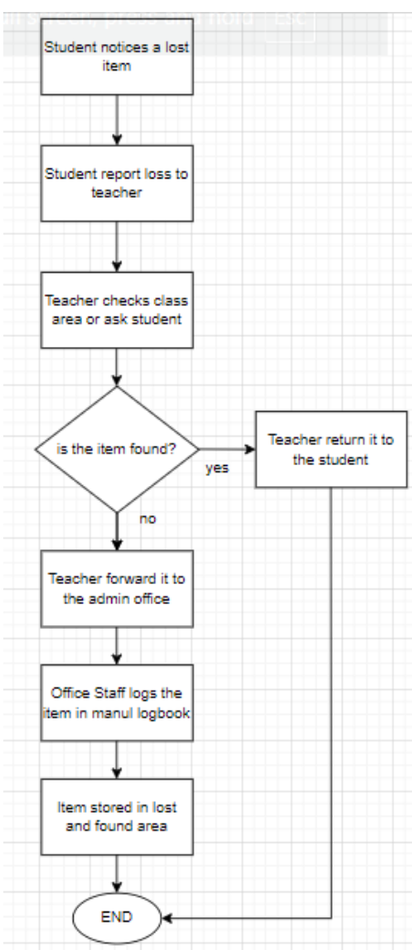
The SWOT analysis evaluates the internal and external factors that affect the proposed Cloud-Based Lost and Found Management System at Pitos National High School. It identifies the system's strengths such as centralized records and automated processes, as well as weaknesses like the need for internet connectivity and training. The analysis also highlights opportunities for future expansion

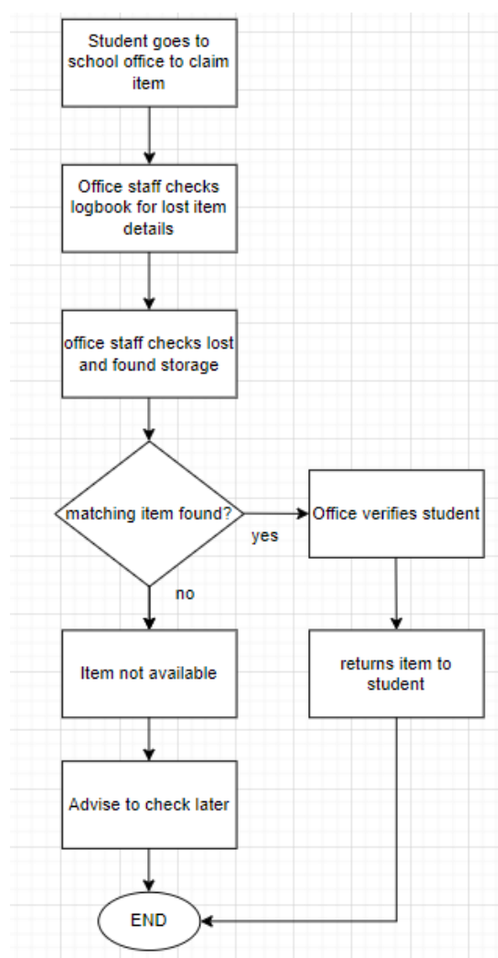
and integration, and threats such as potential system downtime or cybersecurity risks. This assessment helps determine the system's viability and improvement areas.

Strengths	Weaknesses	Opportunities	Threats
Centralized cloud-based database for easy access Fast reporting and claiming process Automated notifications improve communication Secure authentication and controlled access Reduces manual workload for administrators	Requires stable internet connection Training is needed for new users Initial setup costs can be high for a public school Some students may not have mobile data/WiFi access	Can be expanded to include QR item tagging Can be integrated into school portals or DepEd systems Supports school's digital transformation objectives Can become a model system for other public schools	System downtime if cloud service is unavailable Cybersecurity risks if not properly maintained Resistance from users who prefer manual processes ISP outages in public school areas

Current Process Flow

A current process flow is a step-by-step representation of how a manual lost and found report and claiming operation is performed in the existing system.





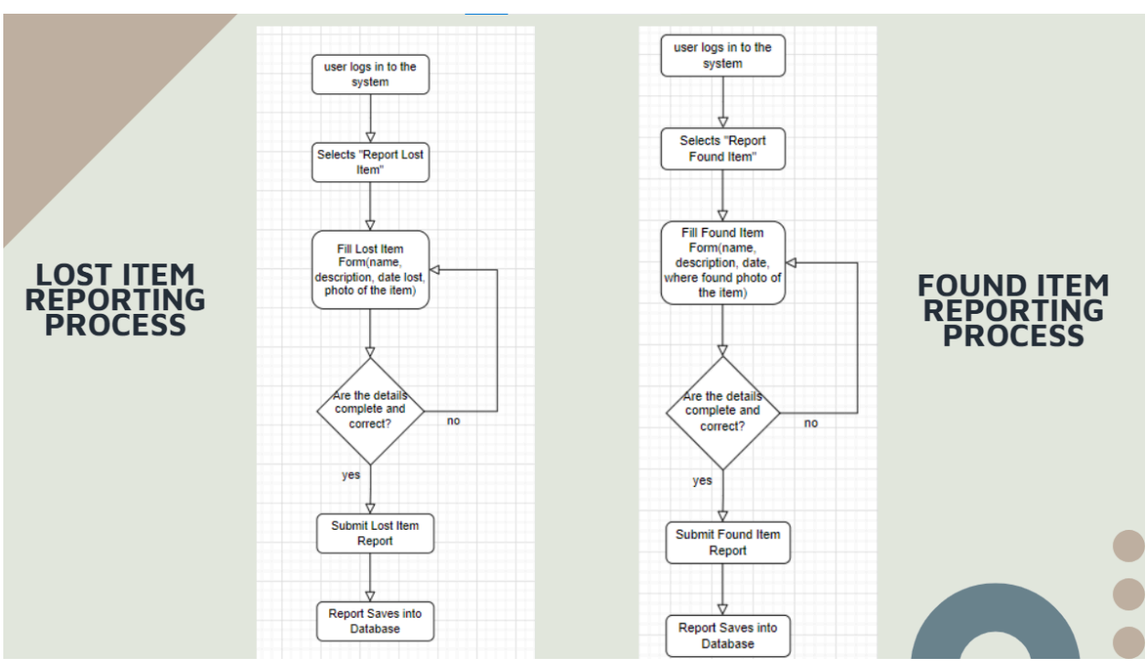
Gap Analysis

The Gap Analysis compares the school's current manual lost and found process with the proposed cloud-based system. It identifies operational gaps such as lack of centralized records, slow item verification, and absence of automated notifications. The analysis shows how the new system will bridge these gaps by introducing digital reporting, real-time updates, organized databases, and improved transparency, resulting in a more efficient and reliable process.

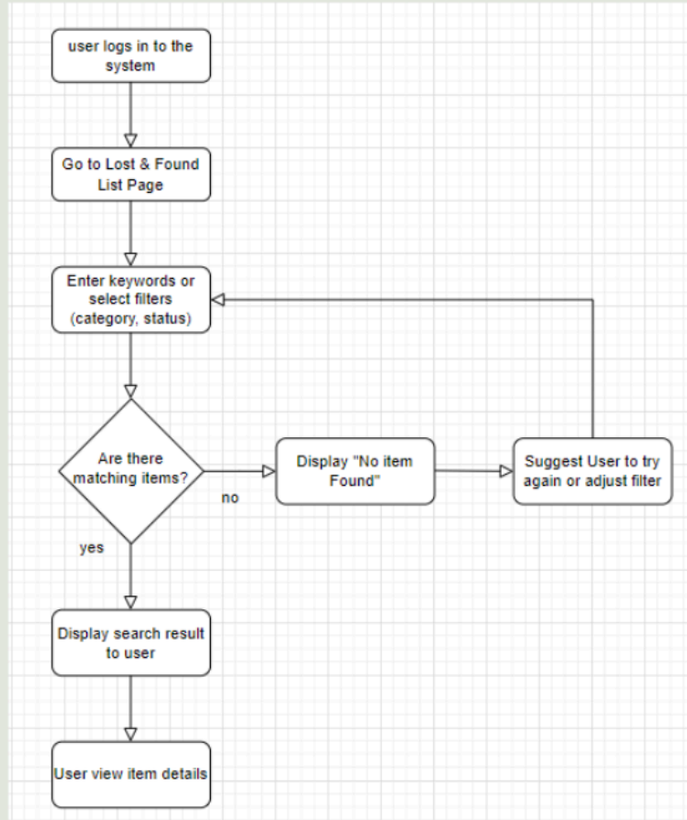
Manual	System	Gaps Identified
Lost and found reporting is verbal or written	Reporting done online through forms	Lack of digital platform
Items often remain unclaimed	Users get notifications and can search database	No automated communication
Hard to verify ownership	Admin can request photos / details digitally	No structured verification
No centralized record	Cloud database stores all item logs	Disorganized and incomplete records
Slow claiming process	Status tracking updates users instantly	No real-time information
No unclaimed item reminders	Automatic alerts for long-unclaimed items	No alert mechanism

Proposed Process Flow

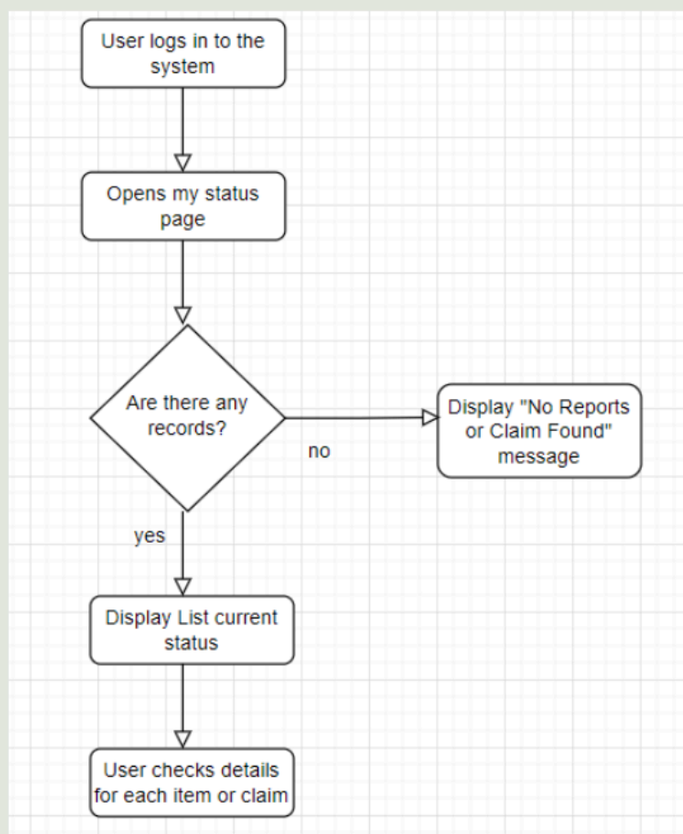
A proposed process flow outlines a redesigned or improved sequence of steps for performing a task or operation. It shows how the process will work after implementing changes, aiming to increase efficiency, reduce errors, and improve overall outcomes compared to the current system.



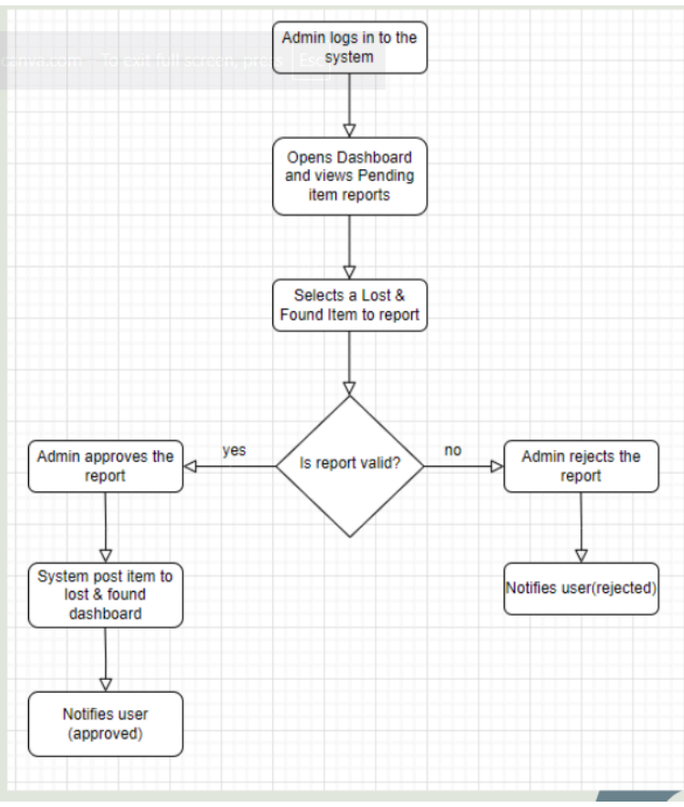
SEARCH & FILTER PROCESS



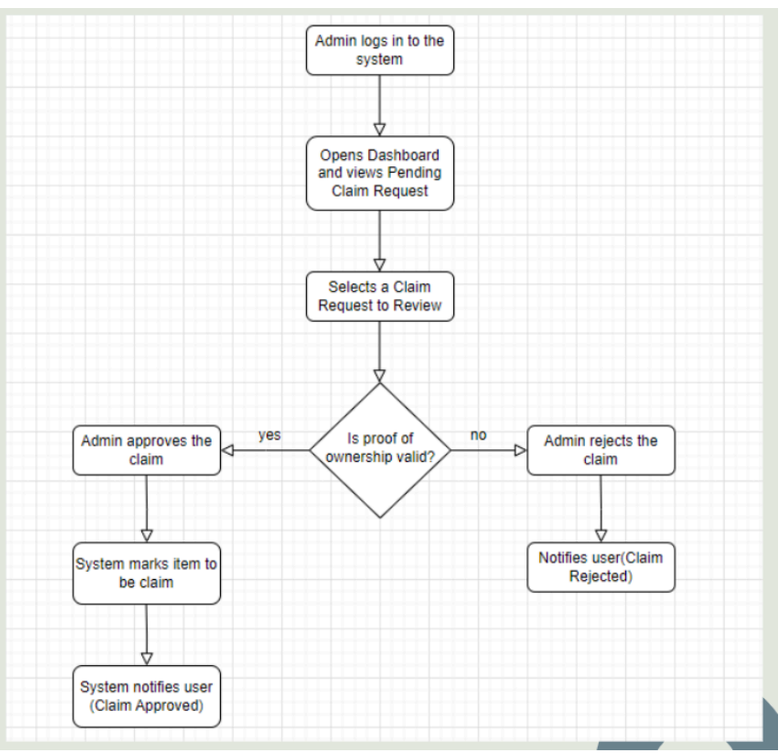
STATUS TRACKING PROCESS

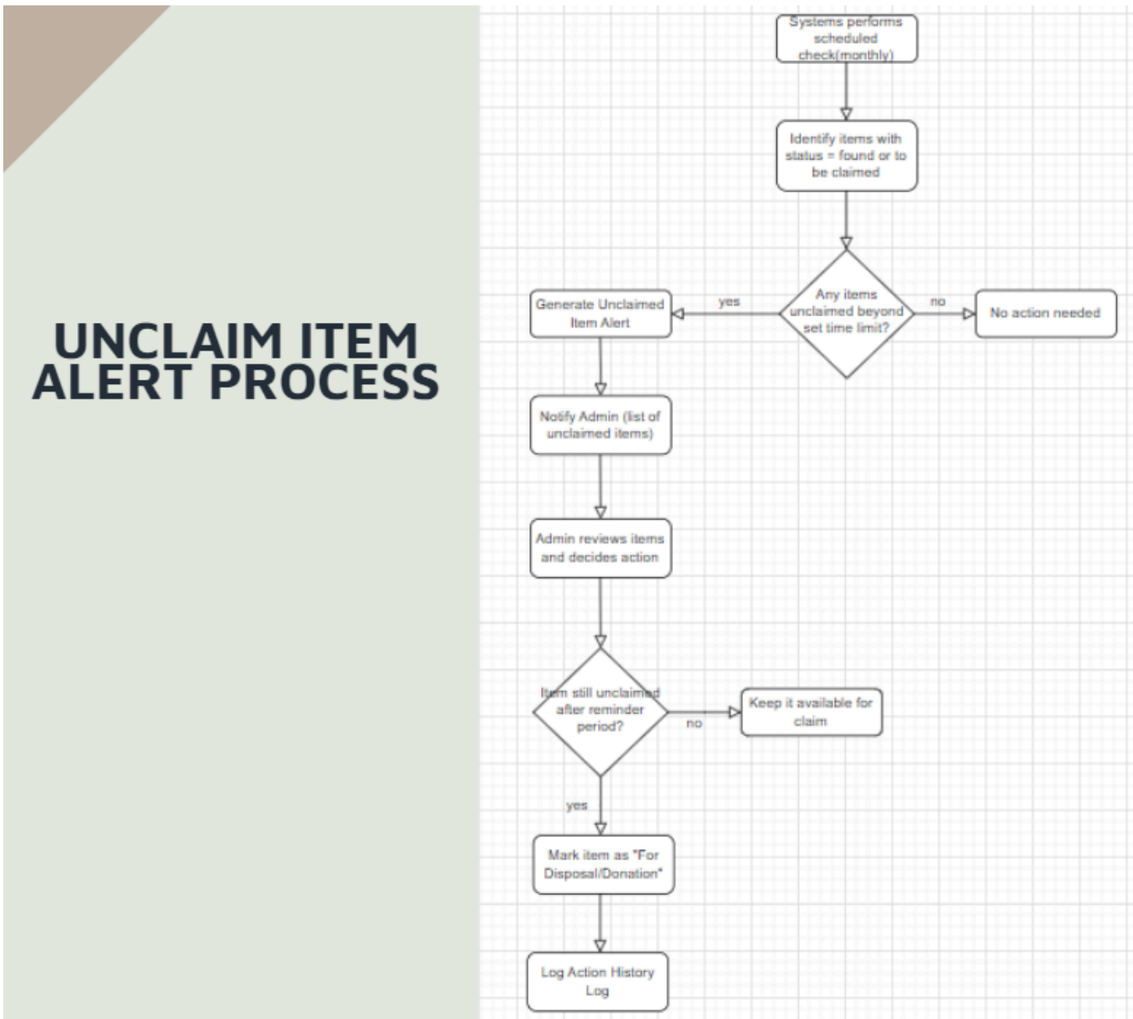


ITEM VERIFICATION & APPROVAL



CLAIM VERIFICATION





Acquisition Strategy

PNHS will implement a Custom In-House Development approach.

Instead of relying on generic off-the-shelf solutions, the school will develop a system tailored specifically to its Lost & Found operations.

Why Custom In-House Development Is the Most Suitable:

- Tailored Functionality: The system will focus solely on the school's Lost & Found processes, avoiding unnecessary features that generic software might include. This ensures staff can use it efficiently with minimal training.
- Full Control and Ownership: By building the system internally, PNHS retains complete control over the software, including the source code and updates, avoiding dependency on external vendors.
- Flexible and Adaptable: Any changes in school policies or procedures can be quickly reflected in the system, allowing updates to be made immediately without waiting for third-party support.
- Secure Data Management: All student and item records will remain under the school's supervision, with secure storage on local or private cloud servers, ensuring confidentiality and reducing security risks.